

STA2020F Tutorial on Logistic Regression: Memo

Question 1

- 1) South Africa is home to 12 species of tortoise, most of which are endangered through habitat loss. CapeNature therefore needs to monitor these species. Different species of tortoises can be told apart based on their appearance. But it is difficult to tell apart the sexes within each species. Researchers collected data on angulate tortoises (*Chersina angulata*) and in this tutorial, we want to see whether we can use morphological measurements to tell whether an individual is female or male. There were 242 tortoises in total. Of these, 99 were females (coded 1) and 143 males (coded 0).
- a) We first used the logistic regression model below to examine whether length (in mm) was a good predictor for sex in this species of tortoise.

ml:

$$Y_i \sim \text{Binom}(1, p_i)$$
$$\text{logit}(p_i) = \beta_0 + \beta_1 \times \text{length}$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	22.72498	2.94934	7.705	1.31e-14
Length	-0.12296	0.01578	-7.793	6.54e-15

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 327.44 on 241 degrees of freedom

Residual deviance: 193.53 on 240 degrees of freedom

AIC: 197.53

- i) What is the effect of length on the log odds of an individual being female (round to two decimal points)?

-0.12

- ii) By what factor do the odds of being a female change for each unit increase in length?

-0.89

- iii) Are longer tortoises relatively more likely to be males or females, compared to short tortoises?

Males. Higher values of length are associated with lower values of log odds of an individual being a female and, therefore, are associated with a lower probability of being a female (or higher probability of being a male). Remember that, in this example female = 1 and male = 0.

- iv) what is the null hypothesis tested in the line starting with 'Length'?

H0: The length of the tortoise has no effect on its probability of being a female.

- v) What do you conclude from the test?

We have strong evidence that length is related to sex.

- vi) Suppose that you have a tortoise of length 200mm. What is the probability that it is a female?

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0.13

- b) From the model in question a), we predicted the sex of each tortoise based on its length. Using a threshold of $p=0.5$, we obtained the following confusion matrix:

pred	obs	
	0	1
0	126	22
1	17	77

- How many tortoises were correctly predicted to be male?
126. Remember that, in this example female = 1 and male = 0.
- For how many tortoises did the model predict sex incorrectly?
39
- The results of the confusion matrix depend on the classification threshold chosen. True / False?
True.
- What is the sensitivity of this classification tool?
 $77/99=0.78$

- c) Now, we fit two more models, m2 and m3 and obtain the following output:

m2:

$$Y_i \sim \text{Binom}(1, p_i)$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 \times \text{length} + \beta_2 \times \text{weight}$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	36.193011	4.588543	7.888	3.08e-15
Weight	0.015496	0.002586	5.992	2.07e-09
Length	-0.280199	0.036273	-7.725	1.12e-14

m3:

$$Y_i \sim \text{Binom}(1, p_i)$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 \times \text{length} + \beta_2 \times \text{weight} + \beta_3 \times \text{length} \times \text{weight}$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.418e+01	1.384e+01	-1.025	0.305542
Weight	7.263e-02	1.733e-02	4.191	2.78e-05
Length	-1.067e-02	7.791e-02	-0.137	0.891063
Weight:Length	-3.031e-04	8.956e-05	-3.385	0.000712

- Based on model m2, what are the log odds of an individual with weight = 900g and length = 180mm to be a female?
-0.30
- Based on model m2, what is the probability of an individual with weight =

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- 900g and length = 180mm to be a female? 0.43
- iii) Based on model m3, what is the probability of an individual with weight = 900g and length = 180mm to be a female?
0.54

Some extra practice

- 1) Students in UCT's STA2020S class were asked if they have ever driven after drinking. They were also asked how many days per month they drink at least two beers. Let p be the probability a student i said "yes" they had driven after drinking modelled against x_i , the number of days per month of drinking at least 2 beers. Regression results are shown below:

Coefficients:

##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-1.5514	0.2661	-5.83	0.000
## DaysBeer	0.19031	0.02946	6.46	0.000

- a) Using the regression output, calculate a, b, c, and d and complete the table below providing the estimated probabilities of ever having driven under the influence (p) for the given days per month of drinking beer.

DaysBeer	4	12	20	28
Predicted p	a	b	c	d

I changed the notation to make it clearer

$$L_a = -1.55 + 0.19 * 4 = -0.79 \Rightarrow p_a = \frac{1}{1 + e^{-L}} = 0.31$$

$$L_b = -1.55 + 0.19 * 12 = 0.73 \Rightarrow p_b = \frac{1}{1 + e^{-L}} = 0.67$$

$$L_c = -1.55 + 0.19 * 20 = 2.25 \Rightarrow p_c = \frac{1}{1 + e^{-L}} = 0.90$$

$$L_d = -1.55 + 0.19 * 28 = 3.77 \Rightarrow p_d = \frac{1}{1 + e^{-L}} = 0.98$$

- b) When DaysBeer is 10, what is the predicted odds of ever driving after drinking?

$$L_{10} = -1.55 + 0.19 * 10 = 0.35 \Rightarrow odds_{10} = e^{L_{10}} = 1.42$$

- c) What is the predicted change in the log-odds of p associated with an increase of 5 days in number of days per month someone drinks more than 2 beers?

$$C5 = 0.19 * 5 = 0.95$$

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- d) Estimate a 95% confidence interval for the odds ratio associated with a unit increase in DaysBeer.

$$\beta \in \hat{\beta}_1 \pm z_{\frac{\alpha}{2}} * se_{\hat{\beta}_1} = 0.19 \pm 1.96 * 0.03 = [0.13, 0.25]$$

$$OR = e^{\beta} \in [e^{0.13}, e^{0.25}] = [1.14, 1.28]$$

- e) Now the sex (coded 1 for male and 0 for female) is included as an x variable along with DaysBeer.

## Coefficients:				
##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-1.7736	0.2945	-6.02	0.000
## DaysBeer	0.18693	0.03004	6.22	0.000
## SexMale	0.6172	0.2954	2.09	0.037

What is the odds ratio of ever having driven after drinking when comparing males to females?

$$e^{0.62} = 1.86$$

- f) What is the predicted probability of ever having driven after drinking for a female who drinks at least 2 beers a day for 15 days of the month?

$$L = -1.77 + 0.19 * 15 = 1.08 \Rightarrow \hat{p} = \frac{1}{1 + e^{-L}} = 0.75$$

- g) What is the predicted probability of ever having driven after drinking for a male who drinks at least 2 beers a day for 10 days of the month?

$$L = -1.77 + 0.19 * 10 + 0.62 = 0.75 \Rightarrow \hat{p} = \frac{1}{1 + e^{-L}} = 0.68$$

- h) Interpret the meaning of intercept given in the model shown in part e? What is the meaning of $e^{\hat{\beta}_0}$ for this model?

$\hat{\beta}_0$ are the log-odds of having driven after drinking for a female who does not have any days of the month for which she drank 2 or more beers per day. $e^{\hat{\beta}_0}$ are the odds or odds-ratio of having driven after drinking for a female who does not have any days of the month for which she drank 2 or more beers per day.

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- 2) An observed binary outcome takes on categories "positive" and "negative" and a logistic regression model is run on an explanatory variable in order to assess how well the variable predicts this outcome.

		Observed	
		Positive	Negative
Predicted	Positive	20	33
	Negative	10	37

The model estimates a predicted probability p and based on the criteria "predicted outcome is positive if p is greater than 0.3", predicted outcomes are assigned. These predictions are compared to the actual observed outcomes as shown in the table above. For this model and this criteria for p , calculate the implied:

- (a) Sensitivity

$$20/(20+10)$$

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## [1] 0.6666667
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- (b) Specificity

$$37/(33+37)$$

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## [1] 0.5285714
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- (c) positive predictive value

$$20/(20+33)$$

```
## [1] 0.3773585
```

- (d) negative predictive value

$$37/(10+37)$$

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## [1] 0.787234
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Question 2

- 3) Outline the following stepwise procedures:
- Forward selection

Type of stepwise procedure that starts with an empty model. Variables are added one by one. At each iteration the variable that provides the best improvement is included.

- Backward elimination

Type of stepwise procedure that starts with the full model. Variables are removed one by one. At each iteration the variable that provides the best improvement is eliminated.

- 4) Discuss two problems with stepwise procedures

Straight from slides:

Stepwise methods will not necessarily produce the best model if there are redundant predictors (common problem). • Models identified by stepwise methods have an inflated risk of capitalising on chance features of the data. They frequently fail when applied to new datasets. They are rarely tested in this way. • Since the interpretation of coefficients in a model depends on the other terms included, it seems unwise to let an automatic algorithm determine the questions we do and do not ask about our data". • Better models and a better understanding of one's data result from focussed data analysis, guided by substantive theory. • The data analyst knows more than the computer, and failure to use that knowledge produces inadequate data analysis.

- 5) Compare and contrast R^2 , Adjusted R^2 , and AIC as criteria for model selection.

R-squared is not useful for model selection. The more complex model will always have a higher R-squared, because whenever you add a variable into the model R-squared increases. Adjusted R-squared accounts for this by penalising model complexity.

The AIC is an estimator of the relative quality of statistical models for a given set of data. Adjusted R-squared is a measure of explained variance in the response variable by the predictors whereas AIC is a trade off between model goodness of fit and complexity that helps to reduce estimated information loss.

- 6) Data is recorded to predict the price of a car based on numerous factors. In total, there are 24 variables. In order to reduce the number of variables in the model, stepwise procedures are considered. Two steps of an algorithm are provided.
- Which stepwise procedure was performed?

Forward selection

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b) What is happening between the first step and the second step?

In the first step, the addition of all the variables that have not yet been included are considered. Stepwise selection reduces the AIC of the model by the largest amount and is, therefore, included. This process is repeated iteratively (i.e. in the second step, the variable that will be included).